

Justin Vinall

Backend-focused software engineer with 15 years of experience building Python and .NET systems on Azure. Designs and delivers APIs, data pipelines, and integrations across application code, databases, and cloud infrastructure—from architecture and implementation through deployment and production troubleshooting.

EXPERIENCE

Avanade, Seattle, WA — Backend Developer, Sr. Consultant

JAN 2018 - PRESENT

Hydraulic Fracturing Analytics | Python, Azure Databricks, SQL Server, GIS

- Led development of an Azure Databricks pipeline that processed data from thousands of wells into geospatial analytics and map layers for Spotfire
- Partnered with petroleum engineers to translate field requirements into simulation and optimization logic that estimated fracture lengths by stage from fluid volume, permeability, and geographic coordinates
- Built spatial-processing workflows with Shapely, GeoPandas, and alphashape to generate and validate fracture polygons, quantify overlapping fracture zones, and identify under-fractured areas
- Designed SQL Server schemas and optimized data access for spatial outputs and summary metrics consumed by Spotfire

Agent Assist | Python, Azure Functions, SQL Server

- Rearchitected a Python Azure Functions application that retrieved customer account data for call-center agents, reducing average response time from 30 seconds to 3 seconds
- Led migration from SAP HANA to SQL Server, translating and validating nearly 5000 lines of SQL and designing indexes for the new schema
- Designed and executed load and performance tests to validate the migrated database and queries against expected production traffic
- Raised code coverage above 90% with Python unit tests and built Azure DevOps YAML pipelines for automated builds, tests, and deployments

Microsoft Partner Network | .NET, ASP.NET MVC, Sitecore, Azure

- Developed full-stack features and reusable Sitecore components using C#, ASP.NET MVC, JavaScript, HTML, and CSS
- Integrated Marketo REST APIs and forms to capture customer data and CTA interactions for analytics and personalization
- Built an Azure Durable Functions workflow using Service Bus to migrate customer data between platforms via application APIs

Vertafore, Bothell, WA — Software Engineer II, Test Automation

SEP 2011 - DEC 2017

- Built UI automation infrastructure with C# and Selenium WebDriver for a new ASP.NET application, mentoring junior engineers through code reviews
- Developed customer database migration and performance-testing tools while supporting production releases through validation, deployment packaging, and application and certificate troubleshooting

EDUCATION

Western Washington University, Bellingham, WA B.S. in Computer Science, Minor in Mathematics

MAR 2011

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SKILLS

Python & Data: Python, Azure Databricks, PySpark, Django, Django REST Framework

Backend Development: C#, .NET, ASP.NET Core, ASP.NET MVC/Web API

APIs & Integration: REST APIs, third-party API integrations, Azure API Management, Azure Service Bus, OpenAPI/Swagger

Microsoft Azure: Functions, App Service, Entra ID, Key Vault, Storage, VNet, Application Insights

Databases: PostgreSQL, SQL Server/T-SQL, MySQL, Cosmos DB, SAP HANA

Containers: Docker, Kubernetes, Azure Kubernetes Service

DevOps & Infrastructure: Azure Pipelines, ARM templates, Bicep

Testing: unit and integration testing, load and performance testing, code coverage

AI-Assisted Development: Claude Code

CERTIFICATIONS

Microsoft Certified: Azure Developer Associate